Code No: 114CS **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, February - 2024
DESIGN AND ANALYSIS OF ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	Define Omega notation.	[2]
b)	Write control abstraction algorithm of divide and conquer method.	[3]
c)	Write an algorithm of Simple Union.	[2]
d)	Explain the properties of Biconnected components.	[3]
e)	State the Principle of optimality.	[2]
f)	Write the applications of Dynamic Programming.	[3]
g)	Explain the properties of LC Branch and Bound.	[2]
h)	Find the sum of subsets for the following set of integers by using fixed tuple method {5,10,15,20,18,12} for W = 30	[3]
i)	Write a non-deterministic algorithm to sort an elements in an array.	[2]
j)	What are the advantages of Non-deterministic algorithms?	[3]

PART - B

(50 Marks)

2.a) Derive the time complexity of Strassen's matrix multiplication.
b) Explain about the Amortized analysis. [5+5]

OR

3.a) Solve the following recurrence equation.
$T(n) = 4T(n/2) + n^2$, $T(0) = 0$
b) Explain about the Binary Search with an example. [5+5]

4.a) Describe about the Biconnected Components with an example.
b) Write an algorithm of DFS and also analyze the time complexity of the same. [5+5]

OR

5.a) Write an algorithm of AND/OR graph.
b) Explain the binary tree traversal with an example. [5+5]

6. Design a three stage system with device types D_1, D_2, D_3. The costs all Rs. 30, Rs.15 and Rs 20 respectively. The cost of the system is to be no more than Rs.105. The reliability of each device type is 0.9, 0.8 and 0.5 respectively. [10]

OR

7.a) Explain about the Job sequencing with deadlines.

b) Write an algorithm of Greedy Knapsack problem and also analyze the time complexity of the same. [5+5]

8. Solve the TSP problem having the following cost-matrix using branch-and-bound technique. [10]

$$\begin{bmatrix} 0 & 10 & 15 & 20 \\ 5 & 0 & 9 & 10 \\ 6 & 13 & 0 & 12 \\ 8 & 8 & 9 & 0 \end{bmatrix}$$

OR

9.a) Write an algorithm of n-queens problem and also analyze the time complexity of the same.

b) Explain about the sum of subsets problem with an example. [5+5]

10.a) Explain graph coloring problem is NP-Hard or NP- complete. Justify your answer.

b) Write a non-deterministic algorithm of sum of subsets problem and also analyze the time complexity of the same [4+6]

OR

11.a) Write a non-deterministic algorithm of 0/1 Knapsack problem and also analyze the time complexity of the same.

b) Differentiate between Deterministic and non-deterministic algorithm. [6+4]

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